When Food Exists but Hunger Persists: Identifying Food Supply–Access Disconnects in U.S. Counties

Website link:https://when-food-exists-but-hunger-persists.base44.app

GitHub link: https://github.com/dnyanadab/DATA-MINING-PROJECT-GROUP-13

Sharon Bayela
University of Colorado Boulder
Center for Community, 2249 Willard Loop Drive,, Suite S355, 123 UCB, Boulder, CO 80309
+1 (303) 434-9749
Sharon.Bayela@colorado.edu

Dnyanada Bhosale
University of Colorado Boulder
Center for Community, 2249 Willard Loop Drive,, Suite S355, 123 UCB, Boulder, CO 80309
+1(720) 382-6295
Dnyanada.Bhosale@colorado.edu

Samanwita Mukherjee
University of Colorado Boulder
Center for Community, 2249 Willard Loop Drive,, Suite S355, 123 UCB, Boulder, CO 80309
+1(720) 655-5627
Samanwita.Mukherjee@colorado.edu

Grant Wiese
University of Colorado Boulder
Center for Community, 2249 Willard Loop Drive,, Suite S355, 123 UCB, Boulder, CO 80309
+1 (614) 288-6933
Grant.Wiese@colorado.edu

ABSTRACT

Food insecurity remains a persistent issue in the United States despite the country's substantial economic wealth and food production capacity. In 2023 alone, approximately 18 million U.S. households experienced food insecurity, highlighting a paradox in which food systems appear robust while many communities still lack reliable access to nutritious food. This research examines the structural mismatch between food infrastructure and actual food access at the county level across the United States. By integrating multiple national datasets, the study constructs two composite indices: the Food Infrastructure Index (FII), representing the density of grocery stores, supercenters, and SNAP-authorized retailers, and the Low Access Indicator (LAI), representing the proportion of low-income and low-access populations. These indices are standardized and used to calculate an Access Mismatch Score that identifies counties where food infrastructure appears sufficient but residents still face significant access barriers. Additional socioeconomic factors such as median household income and vehicle access are incorporated to distinguish between physical access limitations and broader economic disadvantage. The findings aim to reveal geographic patterns of food access inequality and highlight counties where food systems fail despite apparent supply. Ultimately, this research seeks to provide insights that can support policymakers, public health agencies, and community organizations in developing more targeted strategies to address food insecurity.

1. INTRODUCTION AND RELATED WORK

The United States produces and distributes enough food to sustain every resident, yet millions of households continue to face food insecurity. This research reframes the issue not as a failure of supply, but as a structural access problem. By analyzing 3,140 U.S. counties using four distinct analytical methods—regression, classification, association rule

mining, and clustering—this study identifies systemic barriers that prevent physical access to food.

The primary objective is to pinpoint counties where a mismatch exists: areas where food infrastructure appears adequate on paper, but residents remain underserved due to socioeconomic and transportation limitations. Our findings suggest that food access inequality is a deeply rooted structural issue concentrated in the South, the rural Midwest, and remote regions like Alaska.

Previous literature and USDA-defined measures often focus on low-income and low-access conditions as proxies for food insecurity. This study builds upon that foundation by introducing a **Food Infrastructure Index (FII)** and an **Access Mismatch Score** to differentiate between nominal supply and actual accessibility.

Research consistently shows that income, poverty, and unemployment are the strongest predictors of food access outcomes. Our analysis confirms a negative association between median income and low food access.

While SNAP and WIC authorized retailers provide a baseline for assistance, the density of these outlets does not always translate into physical access. Specifically, the presence of convenience stores or SNAP-authorized retailers can create a superficial appearance of supply without addressing barriers like distance.

Recent studies have highlighted the expansion of discount retail in underserved areas. Our models identify Dollar General density as a reliable signal of food system failure, often acting as a lagging indicator of labor market distress rather than a standalone solution for food access.

Documented structural underinvestment in rural and southern regions has created a persistent north-south gradient in food desert risk scores.

2. METHODS , EVALUATION , AND RESULTS

The study utilizes several datasets and analytical techniques to understand these disconnects:

Multiple national datasets were integrated at the county level using **FIPS codes**. These include the **USDA Food Access Research Atlas (2019)** for low-income/low-access variables, the **USDA Food Environment Atlas (2020)** for retail density (grocery stores, supercenters, SNAP retailers), and the **American Community Survey (2015–2019)** for socioeconomic factors like median income and vehicle access.

Two main indices were constructed:

- **Food Infrastructure Index (FII)**: Standardized per capita density of grocery stores, supercenters, and SNAP-authorized retailers.
- **Low Access Indicator (LAI)**: Standardized USDA proxy variables for low-income and low-access populations.

PREDICTIVE MODELLING:

Regression: Used **OLS**, **Ridge**, and **Lasso** regression to establish a baseline relationship between socioeconomic factors and food access.

Classification: Employed **Random Forest**, **CatBoost**, and **XGBoost** to capture non-linear relationships and identify drivers of High Failure food systems.

Association Rule Mining: Applied FP-Tree and Apriori algorithms to discover patterns between high poverty, unemployment, and food system failure.

Clustering: K-Means was used to provide structural context for the data.

EVALUATION

Models were evaluated based on their predictive power and stability:

- **Regression Performance**: All three regression models explained only about 13% of the variance ($R^2 = 0.13$), suggesting that linear models struggle with the complexities of food access data.
- **Model Selection**: Ridge Regression was selected as the best linear model due to its

lowest test **RMSE (0.1741)** and highest cross-validation stability (+/- 0.0821).

- **Classification Metrics**: Recall and precision were the primary metrics for classifiers, with CatBoost emerging as the strongest model due to its high recall and actionable SHAP analysis.
- **Diagnostics**: Residual plots for the regression models revealed heteroscedasticity and a right-skewed distribution, indicating the models were better at predicting typical counties than extreme outliers.

RESULTS

The analysis revealed several key insights regarding food access inequality:

- Rural areas have a significantly higher median low-access share than urban areas.
- High densities of Dollar General stores (per 10k people) are strongly associated (93% confidence) with High Failure food systems. This supports the theory that these stores proliferate in areas lacking traditional grocery access.
- Rural population and grocery store access were identified as top predictors of food inequality through CatBoost's analysis.
- Geographic mapping identified Alaska as a prominent outlier with extreme structural risk, while much of the central U.S. (e.g., Iowa, Nebraska) showed lower risk levels.
- Association rules showed that high unemployment and a lack of vehicle access are nearly certain predictors of a failing food system. Conversely, higher car ownership (Low No-Vehicle Rate) did not necessarily prevent a county from being classified as High Failure.

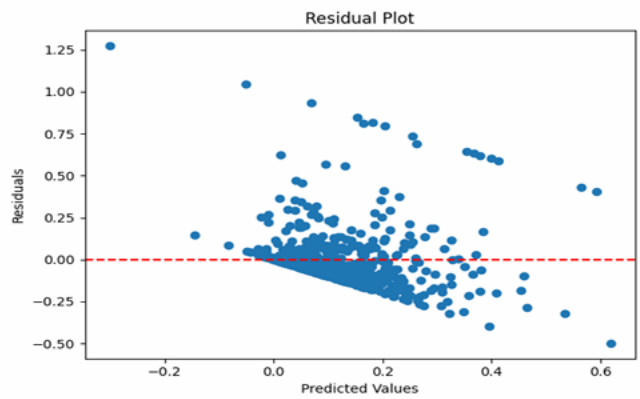

Figure 1 : Residual Plot (Predicted vs. Residuals) — Ridge Model

The residual plot reveals a fan-shaped pattern, with residuals spreading more widely for lower predicted values. This indicates heteroscedasticity, the variance of errors is not constant. The concentration of predictions near zero suggests the model struggles with counties that have very high low-access shares, likely because those extreme cases involve non-linear dynamics not captured by a linear model.

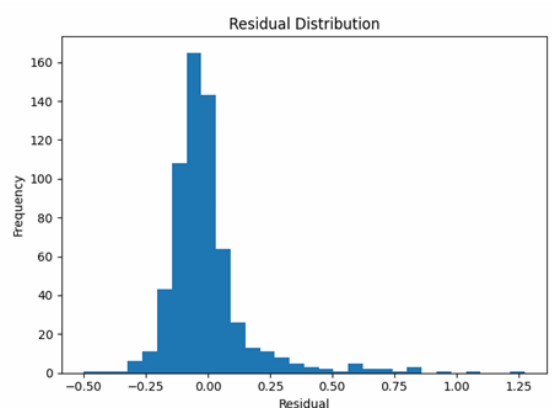

Figure 2 : Residual Distribution — Ridge Model

The residual distribution is right-skewed with a heavy tail, deviating from the normality assumption. Most residuals cluster near zero, but a long positive tail reflects counties where the model significantly underestimates the low-access share.

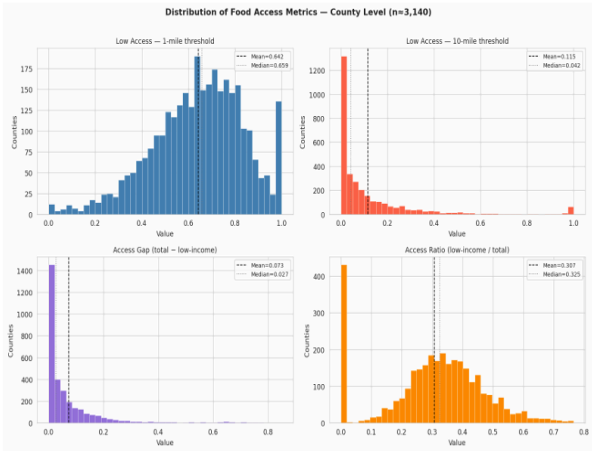

Figure 3: Distribution of food access metrics

For all features used for clustering, the states were created using analysed variables that represent physical access (i.e., low_access_1, low_access_10), economic (PovertyRate and MedianFamilyIncome), coupled vulnerability (low_income_access_10), as well as other engineered variables (like access_gap, access_ratio, DG_per_10k, food_desert_risk), and others. Additionally, robust scaling was used to eliminate the impact of outliers. The data set was reduced in dimension using PCA, while UMAP was only used to produce a visual of the results of clustering.

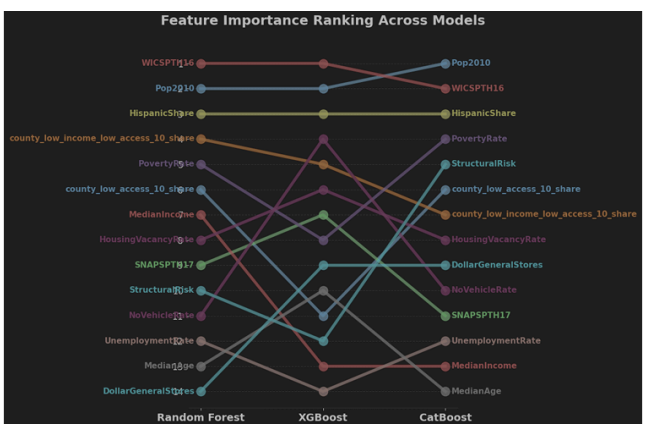

Figure 4 : Feature ranking

Across the classification models (Random Forest, XGBoost, and CatBoost), certain features consistently emerged as the most predictive:

- **Population (Pop2010):** Identified as a top predictor across all models. SHAP analysis further revealed that counties with **extremely low populations** have the least access, aligning with the lack of infrastructure in rural communities.
- **Grocery Store Access (WICSPTH16):** Consistently ranked as a primary driver of food access.
- **Hispanic Share:** Interestingly, this variable appeared in the top three most predictive features, though its exact directionality requires further study of its relationship with other unmeasured variables.

- **CatBoost (Most Predictive Classification Model):** Provided the most actionable insights through **SHAP analysis**. It specifically highlighted Dollar General density as a significant predictor, showing that a higher number of these stores is associated with lower food access, potentially profiting from food deserts.
- **Ridge Regression (Linear Baseline):** Revealed relative importance through standardized coefficients:
    - **Increasing Low Access:** Food Infrastructure Index (0.0450), Retail Proxy (0.0403), and Median Age (0.0281).
    - **Decreasing Low Access:** Median Income (-0.0258), Unemployment Rate (-0.0221), No Vehicle Rate (-0.0204), and Poverty Rate (-0.0193).
- **Features with High Variance:** The rankings for NoVehicleRate, DollarGeneralStores, and MedianIncome changed the most from model to model.

While regression models provided a linear baseline, they only explained about 13% of the variance. Decision tree models like **CatBoost** were preferred for identifying non-linear drivers and practically meaningful findings, such as the correlation between labor market distress (unemployment) and discount retail density.

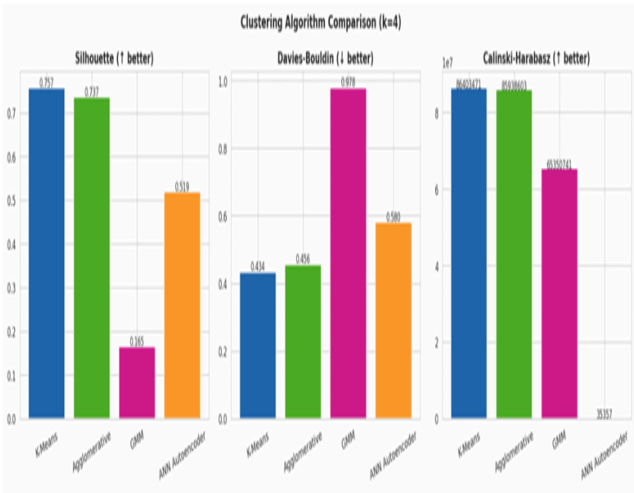

Figure 5: Clustering algorithm comparison

The silhouette score produced by K-Means Clustering was highest at approximately 0.75. There was a good separation within the feature space generated by K-Means. The results generated by Agglomerative Clustering also yielded relatively similar results. On the other hand, GMM and Autoencoders were provided with inferior results due to incorrect assumptions as well as overfitting the dataset.

Method	**Best Model**	**Key Result/Metric**	**Significant Predictors**
Regression	**Ridge**	Test RMSE: 0.1741; R^2 ~0.13	Food Infrastructure, Retail Proxy, Median Age
Classification	**CatBoost**	Highest recall, second-highest precision	Population (Pop2010), Grocery access (WICSPTH16)
Clustering	**K-Means**	Silhouette Score: ~0.75	Socioeconomic variables (Income, Poverty)

Table 1 : Comparison of different models implemented.

3. CONCLUSION AND FUTURE WORK

This study began with a deceptively simple observation: the United States produces and distributes more than enough food to feed every resident, yet millions of households remain food insecure. Rather than treating this as a supply problem, the project reframed it as a structural access problem. Using USDA-defined low-income and low-access conditions as established proxy measures, the research constructed a county-level Food Infrastructure Index (FII) and a Low-Access Indicator (LAI) and calculated an Access Mismatch Score to identify the most troubling cases: counties where food infrastructure appears adequate, yet residents still face significant access barriers.

Across 3,140 U.S. counties and four analytical methods, regression, classification, association rule mining, and clustering, the findings converge on a consistent conclusion: food access inequality in the United States is a structural, socioeconomic problem, not a supply problem. The counties that suffer most are not lacking food in the abstract; they are rural, low-income, and systematically underserved by the transportation and retail networks that make food physically reachable. High-risk counties concentrate in the South, the rural Midwest, and remote regions such as Alaska and New Mexico. Perhaps most tellingly, the density of Dollar General stores whose business model depends on capturing demand in areas abandoned by full-service grocery retail proved to be one of the most reliable signals of a failing food system across every model tested.

Findings by Research Question

Primary Research Question: Which U.S. counties exhibit high levels of low-income/low-access conditions despite strong food availability and infrastructure?

The Access Mismatch Score defined as the standardized difference between the FII and the LAI successfully identified counties in the top quartile of both food infrastructure and low-access indicators as exhibiting structural food access mismatches. Cluster C1 from the K-Means analysis most directly captures this archetype: “Remote Rural Desert Moderate Income” counties, characterized by a low-access share of 0.303 at the 10-mile threshold and a Dollar General density of 0.172 per 10,000 residents, despite having

infrastructure nominally present. Geographically, high-mismatch counties concentrated in the South, rural Midwest, Alaska, and the Southwest regions where the presence of SNAP-authorized retailers or convenience stores creates a superficial appearance of supply without meaningfully addressing access barriers rooted in distance, transportation, and income.

Secondary Research Question 1: How do food infrastructure measures relate to low-access conditions?

The FII components grocery store density (GROCPTH16), supercenter density (SUPERCPTH16), SNAP retailer density (SNAPSPTH17), and WIC outlet density (WICSPTH16) showed significant geographic variation. SNAP retailer density had the highest average per-county value and the widest interquartile range, meaning that in many counties the formal assistance infrastructure exists but does not translate into physical food access. The CatBoost SHAP analysis confirmed that WIC-authorized grocery access (WICSPTH16) was among the two strongest predictors of food system classification, and that its direction is intuitive: higher WIC-authorized store access is associated with better outcomes. However, SNAP retailers and convenience stores, which dominate the retail landscape in underserved counties, do not carry the same protective effect. This suggests that the type of infrastructure matters as much as its density, a finding that challenges policy approaches focused solely on expanding SNAP authorization without attending to the nutritional quality and physical accessibility of authorized outlets.

Secondary Research Question 2: To what degree do socioeconomic and transportation limitations overlap with access mismatches?

The overlap is near-total. Across all four analytical methods, income, poverty rate, and unemployment were the most consistent and strongest predictors of low food access outcomes. The Ridge regression coefficient analysis confirmed that MedianIncome is negatively associated with low food access ***(coefficient: −0.0258)*** and that NoVehicleRate is also inversely associated ***(−0.0204)***, both in the expected direction when controlling for all other variables. The Apriori association rules further quantified the overlap: counties with High Unemployment and High Low Access had a **97.9%** confidence of also being classified as High Low-Income/Low Access. The ACS data revealed that the inverse relationship between median income and vehicle ownership is itself a structural amplifier as income falls, car ownership drops, and transportation barriers compound the distance-based access constraints already measured by the LAI. The Access Mismatch Score was therefore not just a function of missing grocery stores; it was substantially explained by the co-occurrence of economic and mobility deprivation.

Secondary Research Question 3: Do access mismatches cluster geographically?

Yes, with clear and interpretable regional patterns. The K-Means clustering ***(silhouette score: 0.757)*** and the accompanying UMAP visualization confirmed that outlier clusters those with the worst food access profiles concentrate geographically in the rural South, Southwest, and Midwest, while well-served counties dominate the Northeast and upper Midwest. The U.S. Food Desert Risk choropleth showed a consistent north-south gradient in risk scores, with southern counties showing the highest average food desert risk and northeastern states (Connecticut, Rhode Island, New Jersey, Massachusetts, Maryland) anchoring the bottom 15. This geographic pattern validates the construct validity of the Access Mismatch Score: the counties it flags as high-mismatch are not randomly distributed; they reflect decades of documented structural underinvestment in rural and southern food infrastructure. Importantly, the clustering analysis also revealed that most U.S. counties exist along a continuous socioeconomic gradient rather than falling into discrete categories, which has direct implications for how policymakers should design intervention thresholds.

Secondary Research Question 4: What is the relationship between Dollar General store density and food access and income?

The data provide consistent, convergent evidence that Dollar General presence is a reliable indicator of food access failure, though causality cannot be established from observational cross-sectional data. The Ridge coefficient for **DG_per_10k** was ***−0.0137***, indicating a negative association with low food access when controlling for other variables, a pattern that reflects the tendency of these stores to locate in already-underserved markets. The Apriori analysis quantified this more precisely: counties with medium Dollar General density (not even high density) already showed a **96.5%** confidence of food system failure. Counties with high Dollar General density showed a

slightly lower but still striking confidence of 93.2%, and in those cases the rules more frequently included High Unemployment as a co-occurring condition suggesting that beyond a threshold density, Dollar General presence becomes a lagging indicator of labor market distress rather than a standalone food access driver. The CatBoost SHAP analysis confirmed the directional relationship: more Dollar General stores per county is associated with lower food access. While this finding does not establish that Dollar General stores cause food access failure, it is consistent with the hypothesis stated in the project proposal that these stores systematically locate in regions already experiencing structural food insecurity, effectively serving, and potentially entrenching, food desert conditions.

Real-World Impact

These findings have direct and specific implications for three audiences.

For federal and state policymakers, the central message is that food infrastructure investment must be differentiated. SNAP retailer authorization is necessary but not sufficient the type, quality, and geographic reachability of authorized outlets determines whether the infrastructure translates into access. Targeted incentives for full-service grocery retailers to enter high-mismatch counties, combined with transportation subsidies in rural areas, would address the two most structurally important gaps identified by this study.

For public health agencies, the geographic clustering of access mismatches provides a tractable screening tool. The composite risk scores and cluster assignments produced by this study can be used directly to prioritize county-level interventions. The finding that food access follows a continuous gradient rather than a binary food-desert threshold argues for intervention programs with sliding eligibility scales that can serve the large and vulnerable middle of the distribution, not just the most extreme outlier counties.

For community organizations and researchers tracking the role of discount retail in food environments, the Dollar General findings warrant continued scrutiny. The pattern identified here—medium DG density as a near-certain signal of food system failure, high density as a marker of compounding labor market distress—is consistent with reporting that these stores suppress full-service grocery entry in the markets they serve. Longitudinal study of counties before and after Dollar General entry, paired with outcomes data on grocery store openings and closings, would be a high-priority next step.

Limitations, Improvements, and Future Work

Several limitations are inherent in the design of this study and were anticipated in the project proposal.

The temporal mismatch between datasets USDA food access data from 2019 versus food environment infrastructure data from 2020 means the analysis is appropriately interpreted as cross-sectional and structural, not temporal. The analysis does not claim that changes in infrastructure caused changes in access; it identifies co-occurring structural conditions at a single point in time.

County-level aggregation, as noted in the proposal, masks within-county variation. A county classified as mixed-access may contain both well-served urban neighborhoods and severely underserved rural tracts. Future work using census-tract-level analysis would substantially improve the precision of mismatch identification and policy targeting.

The most consequential modeling limitation is the low explanatory power of all three regression models **(test $R^2 \approx 0.13$)**. This is not a failure of model specification it reflects the fact that the structural determinants of food access include factors not captured in any of the four datasets: food affordability and quality at the store level, proximity to employment centers, cultural and language access barriers, hours of operation, and neighborhood safety, among others. The regression results establish a valid linear baseline and identify the direction and relative magnitude of measured predictors, but the large unexplained variance is itself a finding: it confirms that access mismatch is a multi-dimensional problem that publicly available proxy variables can characterize but not fully explain.

Future work should pursue three extensions. First, spatial regression models particularly Geographically Weighted Regression and spatial lag models would account for the spatial autocorrelation between neighboring counties that none of the current models' address. Second, the CatBoost classifier would benefit from a more exhaustive hyperparameter search; the current grid was deliberately limited by computational constraints, meaning the XGBoost comparison is not yet fully valid. Third, and most importantly, the 2019 data baseline should be updated. The post-COVID

food retail landscape marked by accelerated Dollar General expansion, new SNAP online purchasing authorization, and pandemic-era shifts in grocery access

Patterns, represents both a potential disruption to and a natural experiment for the structural patterns identified here.

ACKNOWLEDGMENTS

We would like to extend our deepest gratitude to every member of our team; this project's success is a direct result of their collective hard work and unwavering dedication. We are also profoundly grateful to Dr. Alfonso Bastias for his invaluable guidance and encouragement throughout this process. Finally, we wish to thank the **University of Colorado Boulder's Data Science Department** for providing the academic environment and resources that made this achievement possible.

4. REFERENCES

[1] USDA Economic Research Service. (2019). *Food Access Research Atlas.* https://www.ers.usda.gov/data-products/food-access-research-atlas/

[2] USDA Economic Research Service. (2020). *Food Environment Atlas.* https://www.ers.usda.gov/data-products/food-environment-atlas/

[3] U.S. Census Bureau. (2019). *American Community Survey 5-Year Estimates: DP03, DP04, DP05.* https://data.census.gov

[4] Pareto Software. (2022). *Dollar General Locations Database.* https://simplemaps.com/data/business/dollar-general

[5] Coleman-Jensen, A., et al. (2023). *Household Food Security in the United States in 2023.* USDA ERS Economic Research Report.

[6] Feeding America. (2023). *53 Million People Visited Food Banks.* https://www.feedingamerica.org

[7] SNAP Lifts Families Out of Poverty. (2023). *Center on Budget and Policy Priorities.* https://www.cbpp.org

[8] Chicago Council on Global Affairs. (2022). *Neighborhood Dynamics in SNAP Participation and Food Access.*